



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU.	PR.	CREDITS	EVALUATION		
301BCE24	MATHEMATICS FOR COMPUTER ENGINEERING	2	-	-	2	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] A primary objective is to introduce and develop advanced mathematical skills of students that are imperative for effective understanding of engineering subjects [2] The topics covered will equip them with the techniques to understand advanced level Mathematics and its applications that would enrich logical thinking power.	After completing this course, the student will be able to: CO1: Apply Laplace transform In signal processing, Networking, Machine learning CO2: Apply Fourier series in identifying periodic pattern in time series data CO3: Analyze real world scenarios to recognize when probability is appropriate, formulate problems about the scenarios; creatively model these in order to solve the problems using multiple approaches CO4: Analyze insights from large datasets, data analysis, traffic analysis, error detection

UNIT - I

(08 Hours)

LAPLACE TRANSFORM

Definition, Properties of Laplace transform, Inverse Laplace transform by partial fraction method, Convolution theorem, Simple application of Laplace transform to solve ordinary differential equation

UNIT - II

(07 Hours)

FOURIER SERIES & FOURIER TRANSFORM

Fourier Series: Even and odd functions, Half range expansions. Fourier Transform: Definition and Properties (excluding FFT), Fourier integral theorem, Applications of Fourier transform to solve integral equations.

UNIT – III

(07 Hours)

PROBABILITY

Review of discrete and continuous random variables, Mathematical expectation, Variance and Standard deviation, Moments, Moment generating function, Skewness and Kurtosis, Binomial distribution, Poisson distribution

UNIT - IV

(08 Hours)

STATISTICAL TECHNIQUES

Mean, Median, Quartile, Mode, Mean deviation, Standard deviation, Regression equation of three variables, Multiple correlation and its properties

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

Text/Reference Books:

- 1) Advanced Engineering Mathematics (Wiley), Erwin Kreyzig.
- 2) Higher Engineering Mathematics (Khanna Publishers), B. S. Grewal.
- 3) Advanced Engineering Mathematics (S. Chand), H. K. Dass.
- 4) Applied Mathematics for Engineers and Physicists, L. A. Pipes and L. R. Harville.
- 5) Advanced Mathematics for Engineers, Chandrika Prasad.
- 6) A text book of Engineering Mathematics (Laxmi Publication), N. P. Bali & M. Goyal
- 7) Probability and Statistics (Schaum's Outline Series), Murray Spiegel, John Schiller, R.A. Srinivasan

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III - SEMESTER

COURSE CODE	COURSE NAME	TH.	TU.	PR.	CREDITS	EVALUATION		
302BCE24	OBJECT-ORIENTED PROGRAMMING	03	-	-	03	ESE	CIE	TOTAL
						60	40	100

COURSE OBJECTIVE	COURSE OUTCOME
The objectives of this course is to impart knowledge of : 1. Understand and use the basic programming constructs of C++ 2. Manipulate various C/C++ data types, such as arrays, strings, and pointers 3. Isolate and fix common errors in C++ programs 4. Use memory appropriately, including proper allocation/de-allocation procedures 5. Apply object-oriented approaches to software problems in C++ 6. To learn the essentials of inheritance, file handling, exception handling mechanisms etc. to solve real world problems.	After completing this course, the student will be able to: CO1: Understand the basics of object-oriented programming using C++ CO2: Understand the concept of Classes and Objects CO3: Acquired the knowledge and implement programs using standard design algorithms and flowchart to solve general problems. CO4: Explain programming fundamentals including statements, control flow and array. CO5: Apply the concept of abstraction, polymorphism, inheritance, functions, and classes to design Programs CO6: Illustrate and develop programs using Stream class, File Handling, pointers and exception handling mechanisms to solve real world problems.

Unit I:

(06 Hours)

PRINCIPLES OF OBJECT ORIENTED PROGRAMMING

Procedure Oriented Programming (POP) verses Object Oriented Programming (OOP), Features of Object Oriented Programming, Examples of Object Oriented languages, Applications of OOP, Data types, Declaration of variable, Dynamic initialization of variable, Reference variable, Structure of C++ program, Type casting, control structure.

Unit II:

(06 Hours)

CLASS & OBJECT: Introduction, specifying a class, Access specifier, Defining member functions: Inside class and Outside class, Creating objects, Memory allocations for objects, Special Operators in C++: Scope resolution operator, Memory management operators, Manipulators, Basic Input /Output operators, Array, Arrays of Objects

Unit III:

(07 Hours)

FUNCTIONS IN C++: The Main Function, Function Prototyping, Call by Reference, Return by Reference, Inline Functions, Friend Functions, Static Member Functions.

Constructors & Destructors: Constructors Parameterized Constructors, Copy Constructors, Dynamic Constructors, and Destructors.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

Unit IV:

(4 Hours)

POLYMORPHISM: Introduction of Polymorphism, Types of polymorphism: Compile time Polymorphism, Run time Polymorphism, Operator Overloading, Overloading Operators unary & binary operators, Type Conversions. Function Overloading, Virtual Functions.

Unit V:

(8 Hours)

INHERITANCE & POINTERS IN C++: Inheritance: Derived class, Single, Multiple, Multilevel, Abstract Classes, Virtual and pure virtual functions, Concept of Pointer: Pointer declaration, Pointer operator, Address operator, Pointer arithmetic, Pointer to Object: Pointer to object, 'this' pointer, Pointer to derived class, Array of Pointer

Unit VI:

(8 Hours)

STREAMS & FILES IN C++: Stream classes, stream errors, disk file I/O with streams, File pointers, Error handling in file I/O. File I/O with members functions, overloading the extractions & insertion operators, Memory as a stream object, command-line arguments

Text Book:

- 1) Object Oriented Programming with C++ **E. Balagurusamy**, 8th Edition, 2020, **McGraw Hill**.
- 2) A Tour of C++, by Bjarne Stroustrup, McGraw Hill Publishers 2022.
- 3) Programming: Principles and Practice Using C++ by Bjarne Stroustrup, 3rd Edition, 2024, Addison-Wesley Professional

Reference Books:

- 1) **C++: The Complete Reference** Herbert Schildt 4th Edi. McGraw Hill
- 2) C++ and Object Oriented Programming – Jana, PHI Learning.
- 3) Object Oriented Programming with C++ - Rajiv Sahay, Oxford

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III – SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
303BCE24	OBJECT-ORIENTED PROGRAMMING-LAB	-	-	02	01	ESE	CIE	TOTAL
						25	25	50

COURSE OBJECTIVE	COURSE OUTCOME
The objectives of this course is to impart knowledge of 1. Create and Use Classes and Objects. 2. Develop C++ programs using constructors and destructor. 3. Construct the class diagram for real life problems and implement it using virtual functions, abstract classes. 4. To use pointers in C++ for effective memory management and data manipulation. 5. Develop program in C++ for file handling.	After completing this course, the student will be able to: CO1. Understand the basics of object-oriented programming using C++ CO2. Understand the concept of Classes and Objects CO3. Acquired the knowledge and implement programs using standard design algorithms and flowchart to solve general problems. CO4. Explain programming fundamentals including statements, control flow and array. CO5. Apply the concept of abstraction, polymorphism, inheritance, functions, and classes to design Programs CO6. Illustrate and develop programs using Stream class, File Handling, pointers and exception handling mechanisms to solve real world problems.

Principles of Object Oriented Programming:

1. Study and Installation of Object-Oriented Programming (OOP) in C++

Class & Object:

2. Write programs to show use of classes and objects to define the function inside the class.

3. Write programs using Scope resolution, operator Memory, management operator Manipulator.

Constructors & Destructors:

4. Write a program that initializes values using a parameterized constructor.

Polymorphism:

05. Write programs to overload binary operator using Member function Friend function

Inheritance & Pointers in C++:

06. Write programs to implement multiple inheritances

07. Write programs to implement: Pointer to object, 'this' pointer.

Streams & Files in C++:

08. Write programs to read and write from/to file using Constructor open ()

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

Text Books:

- 1) Object Oriented Programming with C++ **E. Balagurusamy**, 8th Edition, 2020, **McGraw Hill**.
- 2) A Tour of C++, by Bjarne Stroustrup, McGraw Hill Publishers 2022.
- 3) Programming: Principles and Practice Using C++ by Bjarne Stroustrup, 3rd Edition, 2024, Addison-Wesley Professional

Reference Books:

- 1) **C++: The Complete Reference** Herbert Schildt 4th Edi. Mcgraw Hill
- 2) C++ And Object Oriented Programming – Jana, Phi Learning.
- 3) Object Oriented Programming With C++ - Rajiv Sahay, Oxford

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech : III – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU	PR	CREDITS	EVALUATION		
304BCE24-A	WEB PROGRAMMING (Open Elective-I)	3	-	-	3	ESE	CIE	TOTAL
						60	40	100

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] Introduce internet and web development basics, HTML, and essential tools. [2] Enable understanding and application of CSS for web page styling and layout. [3] Equip students with JavaScript for interactive and dynamic web pages.	After completing this course, the student will be able to: CO1: Design well-structured static web pages with HTML, understanding the development environment. CO2: Apply CSS for effective web page presentation and layout. CO3: Implement client-side interactivity using basic JavaScript. CO4: Use JavaScript and DOM for dynamic content and event handling. CO5: Develop basic PHP scripts for form processing and database interaction. CO6: Understand XML/JSON, web services, and apply basic responsive design.

Unit I:

(6 Hours)

Overview of HTML

Introduction: Introduction of HTML, Internet, World Wide Web (WWW), Web Browsers, URL, HTTP, Web Programmers Toolbox.

HTML Common tags: List, Tags, Heading, Paragraph, Styles, Tables, images, forms and UI component, frames, Web site structure.

Unit II:

(7 Hours)

Introduction to CSS

Concept of CSS: Basic Syntax and Structure, Working with CSS: Background Images, Colors, and Properties, Styling Elements: CSS IDs, Classes, Grouping, Width and Height, Borders, Boxes, Margins, and Padding, List Styling and Positioning Using CSS.

Unit III:

(7 Hours)

Client-Side Scripting with JavaScript

Introduction to JavaScript: Syntax, Variables, Data Types, Operators, Control Flow: if/else, switch, Loops (for, while), Functions, Introduction to the Document Object Model (DOM), Accessing and Manipulating HTML Elements using JavaScript.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

Unit IV:

(8 Hours)

JavaScript for Interactivity and DOM Manipulation

Events and Event Handling in JavaScript, Common DOM Events (click, mouseover, submit, change, etc.), Form Handling and Validation using JavaScript, Introduction to the Browser Object Model (BOM): window, location, history, Basic JavaScript for creating dynamic effects.

Unit V:

(8 Hours)

Introduction to Server-Side Scripting (PHP Example) and Databases

Introduction to Server-Side Scripting: Need and Overview.

PHP Basics: Syntax, Variables, Data Types, Operators, Control Structures, Functions, Arrays. Working with HTML Forms in PHP (\$_GET, \$_POST).

Introduction to Databases: Basic Concepts (e.g., MySQL).

Basic Database Interaction using PHP (PDO or MySQLi): Connecting and Simple Queries.

Unit VI:

(6 Hours)

Web Data Formats, Web Services, and Responsive Design

Introduction to XML (Extensible Markup Language): Syntax, Structure, Basic Usage.

Introduction to JSON (JavaScript Object Notation): Syntax, Structure, Basic Usage.

Introduction to Web Services: Concepts of SOAP and REST (brief overview).

Introduction to Responsive Web Design: Viewport Meta Tag, Media Queries.

Text Books:

- 1) **Web Technologies**, Bhupesh Taunk, Aniket Nagane, Third Edition-2023, Nirali Prakashan.
- 2) **Web Technologies**, Dr. Pavan Kumar, First Edition -2023, Lovely Profession university.
- 3) **Web Technology**, N.P. Gopalan & J. Akilandeswari PHI Publisher, Second Edition
- 4) **Web Technologies for Beginners**, Dr. Ashwin Mehta SPD Publisher.

Reference Books :

- 1) **An Introduction to WEB Design and Programming**, Wang-Thomson, Course Technology Inc Publisher.
- 2) **Web Technologies**, Achyut Godbole, Atul Kahate, 3rd edition, 2017, McGraw Hill Publisher.
- 3) **JavaScript: The Complete Reference**, Thomas A Powell, Fritz Schneider, 3rd edition, 2013, Tata McGraw Hill Publisher.
- 4) **HTML & CSS : Complete reference**, Thomas A. puowell, James H. Pence, Fifth Edition, McGraw Hill Publisher

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III - SEMESTER

COURSE CODE	COURSE NAME	TH.	TU.	PR.	CREDITS	EVALUATION		
304BCE24-B	INTRODUCTION TO OBJECT ORIENTED PROGRAMMING (Open Elective -I)	03	-	-	03	ESE	CIE	TOTAL
						60	40	100

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] Understand and use the basic programming constructs of C++. [2] Introduce the fundamental principles of object-oriented programming (OOP). [3] Apply constructors and destructors effectively in real-world applications.	After completing this course, the student will be able to: CO1: Develop simple C++ programs using fundamental constructs such as variables, loops, functions, and conditional statements. CO2: Design and implement classes and objects, demonstrating key OOP concepts. CO3: Implement constructors and destructors to initialize and clean up resources in C++. CO4: Create and demonstrate various types of inheritance to achieve code reusability and hierarchy. CO5: Implement and apply function overloading and operator overloading techniques. CO6: Implement C++ programs that utilize pointers for dynamic memory allocation.

UNIT I

(07 Hours)

INTRODUCTION TO OBJECT ORIENTED PROGRAMMING

Introduction to OOP: Procedure Oriented Programming (POP) versus Object Oriented Programming (OOP), Features of Object Oriented Programming, and Applications of OOP and Data types, Declaration of variable, Structure of C++ program, and control structure.

UNIT II

(06 Hours)

CLASSES & OBJECT

Class & Object: Introduction, Specifying a class, Access specifiers, Defining member functions: Inside class and outside class, creating objects, Memory allocations for objects, Basic Input /Output operators

UNIT III

(07 Hours)

CONSTRUCTORS IN C++

Constructors & Destructors: Constructor, Default Constructor, Parameterized Constructor, Copy Constructors, and Destructors.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

UNIT IV

(07 Hours)

INHERITANCE

Inheritance: Introduction to Inheritance, Base class & Derived class, Types of Inheritance: Single, Multiple, Multilevel, Hierarchical, Hybrid.

UNIT V

(07 Hours)

POLYMORPHISM

Polymorphism: Introduction of Polymorphism, Types of polymorphism: Compile time Polymorphism, Run time Polymorphism, Operator Overloading, Function Overloading.

UNIT VI

(06Hours)

INTRODUCTION TO POINTERS

Pointers in C++: Concept of Pointer: Pointer declaration, Pointer operator, Address operator, 'this' pointer, Pointer to derived class, Array of Pointer

Text Book:

- 1) **Object Oriented Programming with C++**, E. Balagurusamy, 8th Edition 2020, McGraw Hill Publications.
- 2) **Object-Oriented Programming in C++**, Author: Robert Lafore, Publisher: Pearson Education, Edition: 4th Edition, Publication Year: 2001
- 3) **Object Oriented Programming with C++ - A book for Engineering Students**, P. B. Kotur, 8th Edition 2014, Publishers: Sapna Book House (P) Ltd., Bangalore
- 4) **Programming: Principles and Practice Using C++** – Bjarne Stroustrup, Publishers: Addison-Wesley
- 5) **The C++ Programming Language** – Bjarne Stroustrup Publishers: Addison-Wesley
- 6) **Object-Oriented Programming with C++** – David Parsons Publishers: Cengage Learning

Reference Books:

- 4) **C++: The Complete Reference**, 4th Edi. Herbert Schildt **Publisher:** McGraw Hill
- 5) **Object-Oriented Programming with C++** – E. Balagurusamy **Publisher:** McGraw Hill
- 6) **The C++ Programming Language** – Bjarne Stroustrup **Publisher:** Pearson
- 7) **Programming in C++** – Ashok Kamthane **Publisher:** Pearson
- 8) **Object-Oriented Programming in C++** – Robert Lafore **Publisher:** Sams Publishing

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU	PR	CREDITS	EVALUATION		
305BCE24-A	WEB PROGRAMMING– LAB (Open Elective-I)	-	-	02	01	SEE	CIE	TOTAL
						25	25	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] Introduce internet and web development basics, HTML, and essential tools. [2] Enable understanding and application of CSS for web page styling and layout. [3] Equip students with JavaScript for interactive and dynamic web pages.	After completing this course, the student will be able to: CO1: Design well-structured static web pages with HTML, understanding the development environment. CO2: Apply CSS for effective web page presentation and layout. CO3: Implement client-side interactivity using basic JavaScript. CO4: Use JavaScript and DOM for dynamic content and event handling. CO5: Develop basic PHP scripts for form processing and database interaction. CO6: Understand XML/JSON, web services, and apply basic responsive design.

Overview of HTML:

- 1) Write a program to create a profile page to display educational details, Hobbies, Achievement, Your Ideals, etc, using basic tags in HTML
- 2) Write a program to implement HTML code for your own study timetable as shown below. (Hint: use the colspan and rowspan attributes).

HTML Common tags:

- 3) Write a program to design a webpage that makes use of a. Document Structure Tags b. Various Text Formatting Tags c. List Tags d. Image Tags

Concept of CSS:

- 4) Create a web page with one or more frames to improve the layout and make it easier to organize content.
- 5) Create user Student feedback form (use text-box, text area, checkbox, radio button, select box etc.)
- 6) Design a web page of your home town with an attractive background color, text color, an Image, font etc. (use internal CSS).

Introduction to Javascript:

- 7) Develop simple calculator for addition, subtraction, multiplication and division operation using JavaScript
- 8) Create an HTML form with fields for Name, Email, Mobile No, Gender, Favorite Color, and a button. Write JavaScript to display the combined information in a text box when the button is clicked.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

Advance HTML5 and CSS Tags:

- 9) Write HTML/Java scripts to display your CV in navigator
- 10) Write programs using Java script for Web Page to display browsers information.
- 11) Write a javascript program to validate USER LOGIN page.
- 12) Write a javascript program for validating REGISTRATION FORM

Text Books:

- 1) Web Technology, Authored by Poonam Sule and published in 2022
- 2) Ralph Moseley Web Technology Paperback – 1 January 2016, by M.T. Savaliya (Author)
- 3) Joel Sklar, Principles of Web Design, Cengage Learning, 6th Edition, 2015.
- 4) Robert W. Sebesta, Programming the World Wide Web, 7th edition, Pearson Education, 2013

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU	PR	CREDITS	EVALUATION		
305BCE24-B	INTRODUCTION TO OBJECT ORIENTED PROGRAMMING –LAB (Open Elective –I)	-	-	02	01	ESE	CIE	TOTAL
						25	25	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] To understand the fundamental concepts of OOP and set up the C++ development environment. [2] To understand and implement object-oriented programming concepts using classes and objects in C++. [3] To understand the concepts of constructors and destructors in C++ and their role in object lifecycle management.	After completing this course, the student will be able to: CO1: Develop simple C++ programs using fundamental constructs such as variables, loops, functions, and conditional statements. CO2: Design and implement classes and objects, demonstrating key OOP concepts. CO3: Implement constructors and destructors to initialize and clean up resources in C++. CO4: Create and demonstrate various types of inheritance to achieve code reusability and hierarchy. CO5: Implement and apply function overloading and operator overloading techniques. CO6: Implement C++ programs that utilize pointers for dynamic memory allocation.

Write C++ programs for following:

INTRODUCTION TO OBJECT ORIENTED PROGRAMMING

1. Study and Installation of Object-Oriented Programming (OOP) in C++

CLASSES & OBJECT

- Write programs to demonstrate the use of classes and objects
- Write program to demonstrate the use of array of object.

CONSTRUCTORS IN C++

- Write a program that initializes values using a parameterized constructor.
- Write a program to show the order in which destructors are called.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

INHERITANCE

6. Write programs to implement Single inheritance
7. Write programs to implement multiple inheritances
8. Write programs to implement multilevel inheritance
9. Write programs to implement hierarchical inheritance
10. Write programs to implement hybrid inheritance

POLYMORPHISM

11. Write programs to demonstrate the use of function overloading.
12. Write programs to demonstrate the use of operator overloading.

INTRODUCTION TO POINTERS

13. Write programs to implement: Pointer to object, 'this' pointer.
14. Write a function that swaps two numbers using pointers.

Text Book:

- 1) **Object Oriented Programming with C++**, E. Balagurusamy, 8th Edition 2020, McGraw Hill Publications.
- 2) **Object-Oriented Programming in C++**, Author: Robert Lafore, Publisher: Pearson Education, Edition: 4th Edition, Publication Year: 2001
- 3) **Object Oriented Programming with C++ - A book for Engineering Students**, P. B. Kotur, 8th Edition 2014, Publishers: Sapna Book House (P) Ltd., Bangalore
- 4) **Programming: Principles and Practice Using C++** – Bjarne Stroustrup, Publishers: Addison-Wesley
- 5) **The C++ Programming Language** – Bjarne Stroustrup Publishers: Addison-Wesley
- 6) **Object-Oriented Programming with C++** – David Parsons Publishers: Cengage Learning

Reference Books:

- 1) **C++: The Complete Reference**, 4th Edi. Herbert Schildt **Publisher:** McGraw Hill
- 2) **Object-Oriented Programming with C++** – E. Balagurusamy **Publisher:** McGraw Hill
- 3) **The C++ Programming Language** – Bjarne Stroustrup **Publisher:** Pearson
- 4) **Programming in C++** – Ashok Kamthane **Publisher:** Pearson
- 5) **Object-Oriented Programming in C++** – Robert Lafore **Publisher:** Sams Publishing

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III – SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
306BSH24	UNIVERSAL HUMAN VALUES	02	-	-	02	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] To develop of a holistic perspective and harmony [2] To strengthen self-reflection. [3] To develop commitment and courage to act.	After completing this course, the student will be able to: CO1: Recall the need, basic guidelines, and process of value education and describe the human being as a co-existence of self and body. CO2: Explain the concept of harmony within the self and the family, and identify key values such as trust and respect in relationships To understand holistic perception and values in Relationship CO3: Describe the inter-connection of society and nature and explain the vision of a universal human order. CO4: Define professional ethics and explain the role of values in ethical human conduct and decision-making through relevant examples.

UNIT – I

(8 Hours)

Introduction to Human Values

- Need, Basic Guidelines, Content and Process for Value Education
- Self-Exploration – Its Content and Process; Natural Acceptance and Experiential Validation.
- Continuous Happiness and Prosperity – A look at basic Human Aspirations.
- Understanding the Human Being as a Co-existence of Self and Body

(8 Hours)

UNIT – II

Harmony in the Human Being and Family

- Understanding Harmony in the Self – Harmony of Thoughts, Feelings, and Emotions
- Harmony in the Family – the basic unit of human interaction
- Values in Relationships – Trust and Respect
- Understanding the differences between intention and competence

(7 Hours)

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

UNIT - III

Harmony in Society and Nature

- Understanding Society – Institution, Organization, and System
- Vision of a Universal Human Order
- The interconnectedness of society and nature
- Holistic perception of harmony in existence

UNIT – IV

(7 Hours)

Professional Ethics and Application of Values

- Professional Ethics – Definition and importance
- Ethical Human Conduct and the concept of Ethical Competence
- Case studies on value-based leadership, conflict resolution, and sustainable living
- Role of technology in human welfare

Text/Reference Book:

1. "Universal Human Values", Dr. Kuldeep S. Sharma, Books clinic Publishing.
2. "Universal Human Values", Dr. Amol Musale, Eagle Leap Printers & Publishers.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III – SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
307BSH24	CONSTITUTION OF INDIA	02	-	-	02	ESE	CIE	TOTAL
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] Highlight the characteristics of Indian Constitution [2] To understand the structures of governments in India [3] To create a politically aware and responsible citizen	After completing this course, the student will be able to: CO1: Understand the core philosophy of Indian constitution CO2: Understand the structure of Parliament CO3: Understand the structure of state legislature CO4: Understand the role of Election commission of India and bureaucracy

UNIT - I

(7 Hours)

Making of Indian Constitution: Constituent Assembly-Composition, Objectives, Preamble and Salient features of the Indian Constitution, Fundamental Rights, Fundamental Duties, Directive Principles, citizenship act 1955, CAA 2019

UNIT - II

(7 Hours)

Parliament- Loksabha, Rajyasabha, Union Government- structures of union government and functions, President,- Discretionary powers, Emergency Provisions, Prime Minister and Council of Ministers, Executive powers of PM, Judiciary - supreme court of India – executive powers

UNIT - III

(7 Hours)

State Government- Vidhansabha, Vidhanparishad, structures of state government and functions, Governor- Executive powers, Chief Minister- Executive powers, Council of Ministers, Judicial system in states- High court and subordinate courts

UNIT - IV

(6 Hours)

Provisions for Amendments, Federal system- distribution of powers - Union List, the State List, and the Concurrent List, Election Commission of India- Executive powers, bureaucracy- role of bureaucracy

Text Book:

- 1) Durga Das Basu, Introduction to the Constitution of India, Gurgaon; LexisNexis,(23rd edn.)2018.
- 2) M. V. Pylee, India's Constitution, New Delhi; S. Chand Pub.,(16th edn.) 2017.
- 3) J. N. Pandey, The Constitutional Law of India, Allahabad; Central Law Agency,(55th edn.)2018.

Reference Books:

- 1) K. Sharma, Introduction to the Constitution of India, Prentice Hall of India, New Delhi, 2002.
- 2) P.M Bakshi, Constitution of India, Universal Law Publishing House, New Delhi, 1999.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III SEMESTER

COURSE CODE	COURSE NAME	T H	T U	P R	CREDITS	EVALUATION		
308BCE24	COMPUTER NETWORKS	2	-	-	2	ES	CI	TOTA
						E	E	L
						30	20	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] Understand the Fundamentals of Computer Networks [2] Understand the concept of layered models and protocol hierarchies	After completing this course, the student will be able to: CO1: Identify different types of networks (LAN, MAN, WAN, PAN, CAN) and their applications. CO2: Analyze protocol hierarchies and their role in network communication. CO3: Understand IPv4 and IPv6 addressing and analyze their functionalities. CO4: Compare and contrast connection-oriented (TCP) and connectionless (UDP) transport protocols.

Unit I : (6 Hours)

INTRODUCTION TO COMPUTER NETWORKS: Definition, Data Representation, Types of Data flow, Need of Computer Networks, advantages and disadvantages, Goals and Application of Computer Network, Network Hardware Components, Topology with its types, Types of networks: (LAN, MAN, WAN, PAN, CAN), Transmission Impairment, OSI Reference Model, TCP/IP model.

Unit II: (9 Hours)

PHYSICAL & DATA LINK LAYER

Guided media & unguided media, twisted pair coaxial Cable, fiber optics. Wireless transmission, microwave transmission. Introduction: Design Issues with data link layer, Flow Control & Error Control Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ, Sliding Window Protocol, virtual LAN.

Unit III: (7 Hours)

NETWORK LAYER

Services provided to the Transport Layer, Network Layer Design Issues, Switching Techniques, Internet Protocol (IP): IPv4, IPv6, ARP, RARP, DHCP, Routing Protocols -BGP, OSPF, ICMP, IGMP, Traffic Shaping: Leaky Bucket, Token Bucket.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

Unit IV: :

(8 Hours)

TRANSPORT & APPLICATION LAYER

Functions of the Transport Layer, Connection-oriented vs Connectionless services, Open-loop and Closed-loop Congestion Control, Quality of Service (QoS) parameters, Overview of TCP, UDP, HTTP, FTP, DNS

Text Books

- 1) An Engineering Approach to Computer Networking By Keshav S Publisher Pearson Education 2nd Edition
- 2) Computer Networks and Internet By Comer D Publisher Pearson Education 2nd Edition
- 3) Computer Networks" by Andrew S. Tanenbaum
- 4) Data Communications and Networking" by Behrouz A. Forouzan

Reference Books

- 1) Data Communications and Networking By Forouzan TataMcGraw-Hill Publications, 4 th Edition.
- 2) Computer Networking -A Top-Down Approach By James F. Kurose Pearson Publication, 7th Edition.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III – SEMESTER

COURSE CODE	COURSE NAME	TH	TU	PR	CREDITS	EVALUATION		
309BSH24	INTER PERSONAL SKILLS LAB	-	-	02	01	ESE	CIE	TOTAL
						25	25	50

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] To develop effective verbal and non-verbal communication skills for professional and personal interactions. [2] To improve active listening and problem-solving skills for effective decision-making. [3] To foster critical thinking and creative problem-solving for effective interpersonal relationships.	After completing this course, the student will be able to: CO1: Show confidence and effectively manage emotional responses such as stress and stage fear while facing an audience. CO2: Find and correct errors in grammar usage and apply writing skills in essay and report writing. CO3: Understand collaboration, teamwork, and presentation skills through participation in group activities and structured discussions. CO4: Define professional resumes and build online profiles that effectively showcase their qualifications and achievements.

UNIT - I

(19 Hours)

COMMUNICATION AND SPEECH DEVELOPMENT (FUNDAMENTAL LEVEL)

Prerequisites: Basic Language Proficiency, Listening Skills, Teamwork and Collaboration Skills

Communication Development: Emotional Development (facing the audience, overcoming consciousness, stress test, confidence building) Introduction, Point of View (Scripted) - Basic Level, Point of View (w/o Script) - Basic Level.

Task 1: Short Activities (1 minute) . Example: Each participant delivers a 1-minute self-introduction in front of a group.

Task 2: Prepare a 2-minute speech on a given topic

Task 3: Impromptu speech on a random topic (e.g., "Your Favorite Hobby") to build spontaneity and quick thinking.

Speech Development: Speech Training (Clarity, Fluency, Rhythm Pronunciation and Speed building), Pronunciation Training

Task 4: Speech Fluency Test. Example: Practice tongue twisters (e.g., "She sells seashells by the seashore")

Task 5: Pronunciation Test. Example: Recite short paragraphs aloud while focusing on natural pauses and rhythm to develop fluency.

Task 6: Speed-reading sessions with comprehension questions to balance reading speed and understanding.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

UNIT - II

(13 Hours)

BASICS OF ENGLISH, VERBAL ABILITY & WRITTEN ENGLISH TEST [WET]

Fundamentals: Vocabulary, Grammar, Verbal and Written skills), Fundamental Vocabulary Development, Fundamentals of English Grammar, Grammar Usage, Error Detection and Correction.

Writing: Essay Writing, Report Writing.

Task 7: Vocabulary Development Games. Example: Word association game to build vocabulary (e.g., "Synonym Challenge" or "Hangman with New Words").

Task 8: Grammar Error Detection Exercise. Example: Provide sentences with deliberate grammatical errors for participants to identify and correct.

Task 9: Essay Writing Practice. Example: Write a 300-word essay on a current topic (e.g., "Impact of Social Media on Youth").

Task 10: Report Writing Task. Example: Write a business report on a mock project (e.g., "Sales Performance Report for Q1 2025").

UNIT - III

(12 Hours)

COMMUNICATION AND PRESENTATION DEVELOPMENT

Task 11: Group Activity – I (Team Introduction) Example: Each group introduces its members creatively in front of the class.

Task 12: Group Activity – II (Collaborative Problem-Solving) Example: Groups discuss and present solutions to a given real-world problem (e.g., "Reducing Plastic Waste on Campus").

Task 13: Point of View :- (Structured Debate) Example: Debate on a topic like "Technology in Education: Boon or Bane?"

Task 14: Presentation Practice: - Create a 5-minute presentation on any assigned topic using PowerPoint and present it to the group.

UNIT - IV

(16 Hours)

RESUME & PROFILE BUILDER AND MOCKS

Task 15: Resume Making – I: - Resume Building Workshop Example: Participants draft their resumes based on a provided template, followed by peer review and feedback.

Task 16: Profile Builder – II: - Create a LinkedIn profile with optimized headlines, summaries, and professional details.

Task 17: Mock Resume Evaluation Example: Participants exchange resumes and evaluate each other's resumes against given criteria.

Task 18: Mock Written Test Example: Conduct a mock company-specific written test covering verbal ability, reasoning, and grammar skills.

Task 19: Group Discussion (Elimination Round) Example: Simulate a GD with elimination based on communication, logical reasoning, and participation.

Task 20: Mock Personal Interview Example: Conduct mock interviews with feedback on communication, body language, and answers.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

Text Book:

- 1) "Communication Skills" by Sanjay Kumar & Pushp Lata – Oxford University Press
- 2) "English Grammar in Use" by Raymond Murphy – Cambridge University Press
- 3) "Effective Technical Communication" by M. Ashraf Rizvi – Tata McGraw-Hill
- 4) "Speak with Power and Confidence" by Patrick Collins – Random House
- 5) "The Elements of Style" by William Strunk Jr. and E.B. White – Macmillan

Reference Books:

- 1) "Developing Communication Skills" by Krishna Mohan & Meera Banerjee – Macmillan India Ltd.
- 2) "Public Speaking for Success" by Dale Carnegie – HarperCollins
- 3) "Vocabulary for Competitive Examinations" by R.S. Aggarwal – S. Chand Publishing
- 4) "Business Communication: Process and Product" by Mary Ellen Guffey – Cengage Learning
- 5) "Pronunciation Practice Activities" by Martin Hewings – Cambridge University Press

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

B.Tech: III – SEMESTER

COURSE CODE	COURSE NAME	TH.	TU	PR	CREDITS	EVALUATION		
310BCE24	FUNDAMENTALS OF PYTHON PROGRAMMING (MDM-I)	-	-	04	02	ESE	CIE	TOTAL
						50	50	100

COURSE OBJECTIVE	COURSE OUTCOME
The Course aims to: [1] To understand the basic syntax, structure, and data types of Python. [2] To develop problem-solving skills using Python programming concept	After completing this course, the student will be able to: CO1: Understand the basics of Python programming for data analytics. CO2: Demonstrate with NumPy for data manipulation. CO3: Illustrate with Pandas for data Preprocessing & manipulation. CO4: Interpret Visualization and analyze data using Matplotlib.

UNIT - I

(17 Hours)

PYTHON BASICS FOR DATA ANALYTICS

Prerequisites Basics of C/C++ Programming, Introduction to Python for Electronics Data Processing, **Python Environment Setup** Jupyter Notebook, Anaconda, Google Colab, **Python Fundamentals** Variables, Data Types, Operators, **Control Flow** if-else, loops, functions, **Data Structures** Lists, Tuples, Dictionaries and Sets, **Working with Files** Working with CSV and Excel files using Pandas,

Task 1: Introduction to Python, Installation of Jupyter Notebook/Anaconda.

Task 2: Write a Python script to read sensor data from a CSV file and display it.

Task 3: Implement a Python program to calculate basic mathematical operations on signal data.

UNIT - II

(04 Hours)

NUMPY BASICS

Introduction to NumPy Arrays, Indexing, Slicing, Mathematical and Statistical Operations using NumPy, Reshaping and Broadcasting.

Task 4: Write a program to perform matrix operations using NumPy for signal processing.

UNIT - III

(19 Hours)

DATA HANDLING & PROCESSING WITH PANDAS

Introduction to Pandas Series and DataFrame Objects, **Data Cleaning** Handling Missing Values, Duplicates, Data Aggregation and Grouping, Merging and Joining DataFrame.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027



National Rural Development and Research Society's

Suryodaya College of Engineering & Technology

Survey No. 117/1 & 117/3 A, Near Dighori Naka, Vihirgaon, Umrer Road, Nagpur, Maharashtra 441204.

(An Autonomous Institute Affiliated to RTM Nagpur University, Nagpur)

Accredited with Grade "A+" by NAAC

DEPARTMENT OF COMPUTER ENGINEERING



DIPLOMA PROGRAM
CE & ME

Task 5: Load an IoT dataset using Pandas and perform basic data exploration.

Task 6: Implement missing value handling techniques in a dataset using Pandas.

Task 7: Generate a line graph using Matplotlib for temperature sensor readings over time.

Task 8: Create a histogram of voltage levels recorded in an IoT-based circuit.

Task 9: Create a scatter plot to visualize frequency vs amplitude data of a signal.

Task 10: Perform data aggregation on a dataset of communication signal strengths.

Task 11: Plot and analyze the trend of an electronics component failure dataset.

Task 12: Implement a moving average filter on noisy sensor data using Pandas.

UNIT - IV

(14 Hours)

DATA VISUALIZATION USING MATPLOTLIB

Basics of Matplotlib Figures, Axes, Labels, Titles, **Matplotlib** Line Plots, Bar Charts, Histograms, Scatter Plots. Customizing Plots Colors, Markers, Annotations, **Applications** Real-world Applications in Electronics Data

Visualization, Signal Data Analysis, IoT Sensor IoT Sensor Data Visualization and Frequency Response Graph.

Task 13: Create a subplot layout to compare multiple sensor readings in an IoT system.

Task 14: Create a real-time data visualization dashboard using Matplotlib.

Mini-project: Data Analysis on an IoT or Communication Dataset.

(06 Hours)

Text Book:

- 1) "Python Crash Course" by Eric Matthes
- 2) A beginner-friendly introduction to Python, covering syntax, control flow, functions, and more.
- 3) "Automate the Boring Stuff with Python" by Al Sweigart
- 4) Focuses on practical Python applications, including automation and data processing.
- 5) "Python Programming: An Introduction to Computer Science" by John Zelle
- 6) Great for learning the fundamentals of Python programming.
- 7) "Python Data Science Handbook" by Jake VanderPlas

Websites & Online Resources:

- 1) Python Official Documentation - <https://docs.python.org/3/>
- 2) Official guide for Python basics, control flow, functions, and modules.
- 3) W3Schools (Python Tutorials) - <https://www.w3schools.com/python/>
- 4) Beginner-friendly tutorials with interactive examples.
- 5) GeeksforGeeks (Python) - <https://www.geeksforgeeks.org/python-programming-language/>

YouTube Channels for Learning Python & Data Science

- 1) Programming with Mosh – Great for beginners (Python Course)
- 2) Corey Schafer – Covers Python, Pandas, Matplotlib, and more.
- 3) Data School – Specializes in Pandas and data science tutorials.
- 4) StatQuest with Josh Starmer – Excellent for statistics and visualization.

BOS Chairman	Dean Academics	Principal	Version 1.0	Applicable for 2024-2027